

PHISHGUARD

ML-Powered Phishing URL Detection System
with Cyber Threat Intelligence Integration

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96.47%	99.44%	3,033	100K	5
MODEL ACCURACY	ROC-AUC	FEATURES	TRAINING URLS	API ENDPOINTS

1. INTRODUCTION

Phishing attacks remain one of the most prevalent cyber threats, frequently bypassing traditional static blacklists by utilising newly registered domains and obfuscated URLs. PhishGuard addresses this vulnerability by combining statistical Machine Learning classification with active Cyber Threat Intelligence (CTI), mirroring real-world Security Operations Center (SOC) workflows.

The system operates as a real-time Chrome browser extension that analyses visited URLs on the fly. It decodes obfuscation, queries live threat feeds (VirusTotal), and synthesises results into structured threat reports accessible via an analyst dashboard.

2. SYSTEM ARCHITECTURE

PhishGuard is composed of four loosely coupled layers:

Layer	Component	Technology
ML Pipeline	Feature extraction, model training, serialisation	Python, XGBoost, scikit-learn, TF-IDF
Backend API	REST endpoints, inference orchestration, CTI lookup	FastAPI, Uvicorn, joblib, VirusTotal API
Browser Extension	Real-time scanning, badge, warning banner	Manifest V3, JS, Chrome APIs
Analyst Dashboard	Scan history, charts, threat reports, CSV export	React 18, Recharts, plain HTML

3. MACHINE LEARNING PIPELINE

3.1 Dataset

The model was trained on the Phishing and Legitimate URLs dataset (Kaggle, harisudhan411), containing 822,010 labelled URLs balanced to 100,000 samples (50,000 phishing, 50,000 legitimate). The dataset provides a modern, diverse collection of real phishing URLs with clearly distinguishable lexical patterns.

3.2 Feature Engineering

Two complementary feature sets are extracted from each URL:

Handcrafted features (33): Lexical and structural properties computed deterministically — no external API calls, sub-millisecond extraction.

FEATURE	DESCRIPTION	TYPE
<code>url_length</code>	Total character length of the URL	Length
<code>domain_length</code>	Character length of the hostname	Length
<code>num_dots</code>	Number of dots in hostname	Count
<code>num_hyphens_domain</code>	Hyphens in domain core only	Count
<code>num_subdomains</code>	Number of subdomain levels	Count
<code>has_ip_address</code>	IP used instead of domain name	Boolean
<code>has_https</code>	Protocol is HTTPS	Boolean
<code>has_prefix_suffix</code>	Hyphen present in domain core	Boolean
<code>has_suspicious_tld</code>	TLD in known phishing set (.tk, .xyz, etc.)	Boolean
<code>has_phish_keyword</code>	Suspicious keywords: login, verify, secure...	Boolean
<code>is_trusted_domain</code>	Domain in curated whitelist	Boolean
<code>url_entropy</code>	Shannon entropy of hostname	Float
<code>domain_entropy</code>	Shannon entropy of domain core	Float
<code>consonant_ratio</code>	Ratio of consonants (random domains score high)	Float
<code>min_brand_levenshtein</code>	Min edit distance to top 24 brand names	Integer
<code>brand_levenshtein_ratio</code>	Normalised brand edit distance	Float

TF-IDF character n-gram features (3,000): The raw URL string is vectorised using character-level bigrams through 4-grams (analyzer='char_wb', ngram_range=(2,4), max_features=3000, sublinear_tf=True). This automatically learns suspicious sequences such as '-secure-', '.php?id=', '//login', 'wp-content/uploads', and obfuscated brand names like 'paypa1' or 'g00gle'.

3.3 Model

An XGBoost classifier is trained on the combined feature matrix (3,033 dimensions per URL). Handcrafted features are StandardScaler-normalised before concatenation with the sparse TF-IDF matrix using `scipy.sparse.hstack`. The TF-IDF vectoriser is fit exclusively on training URLs to prevent data leakage.

Hyperparameter	Value	Rationale
<code>n_estimators</code>	200	Balances accuracy and training time

max_depth	7	Captures feature interactions
learning_rate	0.05	Conservative rate with more trees
subsample	0.8	Reduces overfitting
colsample_bytree	0.6	Lower due to wide feature space
reg_alpha	0.1	L1 regularisation
reg_lambda	1.5	L2 regularisation
tree_method	hist	Fast histogram method for sparse matrices

4. EVALUATION RESULTS

96.47%	99.44%	97%	95%	96%
ACCURACY	ROC-AUC	PHISHING PRECISION	PHISHING RECALL	LEGIT PRECISION

The model was evaluated on a held-out test set of 20,000 URLs (80/20 stratified split). The 96.47% accuracy clears the project requirement of $\geq 95\%$. The ROC-AUC of 0.9944 indicates excellent discriminative ability across all classification thresholds. A trusted-domain whitelist further eliminates false positives on well-known legitimate sites (Google, GitHub, Amazon, etc.).

Class	Precision	Recall	F1-Score	Support
Legit	0.96	0.97	0.97	10,000
Phishing	0.97	0.95	0.96	10,000
Accuracy			0.9647	20,000

5. REST API BACKEND

The FastAPI backend orchestrates URL feature extraction, model inference, and live CTI lookups. Models are loaded once at startup into memory for sub-500ms response times.

Method	Endpoint	Description
GET	/health	Health check and API status
GET	/stats	Model metrics and scan statistics
POST	/scan	Scan a single URL — returns verdict + confidence
POST	/scan/batch	Scan up to 100 URLs in one request
GET	/history	Recent scan history (last 500 scans)

6. CYBER THREAT INTELLIGENCE INTEGRATION

PhishGuard integrates with the VirusTotal API to augment ML predictions with crowd-sourced threat intelligence. For each scanned URL, the system queries VirusTotal's /urls endpoint and retrieves verdicts from 70+ security vendors. The final risk score combines both the ML model confidence and the VirusTotal malicious/suspicious vendor count, providing defence-in-depth against novel phishing domains not yet learnable from lexical features alone.

7. CHROME BROWSER EXTENSION

The Manifest V3 extension provides real-time protection directly in the browser:

- **popup.js** — On icon click, scans the current tab URL via POST /scan and renders a verdict card with confidence bar.
 - **background.js** — Service worker that automatically scans every completed navigation and updates the toolbar badge: ✓ (safe) / ! (medium) / ✗ (phishing).
 - **content.js** — Injects a red warning banner at the top of any page scoring HIGH risk, with a dismiss button.
 - **Storage** — Scan history and daily count persisted in chrome.storage.local.
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8. ANALYST DASHBOARD

A single-file React 18 dashboard (no build step) provides full visibility into scan activity. Features include:

- Live stats: total scans, phishing detected, safe rate, model accuracy
 - Risk distribution pie chart and risk-level bar chart (Recharts)
 - Confidence timeline area chart for the last 20 scans
 - Full scan history table with confidence bars and risk badges
 - Manual URL scanner with inline threat report
 - One-click CSV export of all scan records
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9. TECHNOLOGY STACK

Domain	Libraries / Tools
Machine Learning	Python 3.12, XGBoost, scikit-learn, scipy, Pandas, NumPy
NLP / Features	TF-IDF (sklearn), custom Levenshtein, Shannon entropy
Backend API	FastAPI, Uvicorn, Pydantic v2, joblib
Threat Intel	VirusTotal API v3
Browser Extension	JavaScript (ES2022), Chrome Manifest V3, WebExtensions API
Dashboard	React 18, Recharts, JetBrains Mono, Orbitron

10. CONCLUSION

PhishGuard successfully demonstrates that combining handcrafted URL features with learned character n-gram representations produces a robust phishing detector that exceeds the 95% accuracy target (96.47%) while maintaining sub-500ms inference latency. The two-stage classification pipeline — whitelist fast-path followed by ML inference — eliminates false positives on well-known legitimate domains without compromising detection of novel phishing URLs.

The integration of VirusTotal CTI provides defence-in-depth for domains that may evade purely lexical analysis. The Chrome extension and analyst dashboard deliver this capability directly to end-users and security analysts in a production-ready interface, completing a full end-to-end threat detection system.